

DETERMINATION OF FREQUENCY OF AN ELECTRICALLY VIBRATING TUNING FORK – MELDE’S METHOD

Aim:

To determine the frequency of a vibrating tuning fork using Melde’s arrangement

Apparatus:

Storage cell eliminator, plug keys, connecting wires, Meter scale, thread, weight-end electrically maintained tuning fork.

Description:

An electrically maintained tuning fork consists of electro magnet between the two prongs of tuning fork without touching either of the prongs as shown in figure. To one of the prongs a thin brass plate with an adjustable screw is riveted on it. By adjusting the screw, contact is established with the thin brass plate.

Electrical connections are made as shown in figure 1. When plug is inserted the circuit is completed and electrical current flows through the circuit. Energizing the electromagnet by pulling both the prongs in inward, the circuit is broken immediately at the points, the electromagnet loses its magnetism and the prongs fly back to its original position. Consequently contact is once again established at the circuit and the process repeats automatically.

One end of the thin thread is connected to a small screw provided on one of the prongs of the tuning fork. The other end of the thread is connected to a light card board pan and the material of the thread is passed on a small friction free pulley fixed on to a stand kept at a distance of 3 to 4 m from the fork. Small weight is placed on card board pan so that sufficient tension is created to the string. The tension in the string can be attached by changing the weights in the card board pan. The tuning fork is arranged as shown in figure.

Procedure:

The apparatus is first arranged for transverse mode of vibrations with length of the string 30 to 40 cm and passing it over the pulley, the electric circuit is closed and the rheostat is adjusted till the fork vibrates steadily. The load in the pan is adjusted slowly, till a convenient number of loops with defined nodes and maximum amplitude at the antinodes are formed. The vibrations of string are being in the vertical plane.

The no. of loops (p) formed in the string between pulley and the fork is noted. The length of the string (l) between the pulley and the fork is noted..

For longitudinal vibrations of the prong are parallel, to the length of the string. After noting the observations in this position, the tuning fork is arranged as shown in figure (2). For transverse vibration, the vibrations of string should be perpendicular to the length of the string. It is to be noted that in both cases, the stretched string should vibrate at right angles to its length. In case the stretched string after forming loop rotates, then

small weights are either to be added into the pan or removed to create sufficient tension in the string and the loops formation is verified.

Theory:

A. Transverse Arrangement:

When the direction of motion of the prong is at right angles to the length of the string, the vibrations of the thread represent the transverse mode of vibration.

The fork is placed in transverse vibration position and by adjusting the length of the string, weights in the pan. The string starts vibrating with a frequency exactly half the frequency of the fork.

During longitudinal vibrations, when the prong in its right extreme position the string corresponding to a loop gets slackened and comes down and when the fork is in its left extreme position the string corresponding to the loop gets slackened and comes down string moves up to its initial horizontal position and becomes light.

But when the prong is again in its right extreme position there by completing one vibration, the string goes up, its inertia carrying its upward and there by completes only a half vibration. Hence the frequency of each loop is

$$n = \frac{p}{2l} \sqrt{\frac{T}{\mu}} \text{----- (1)}$$

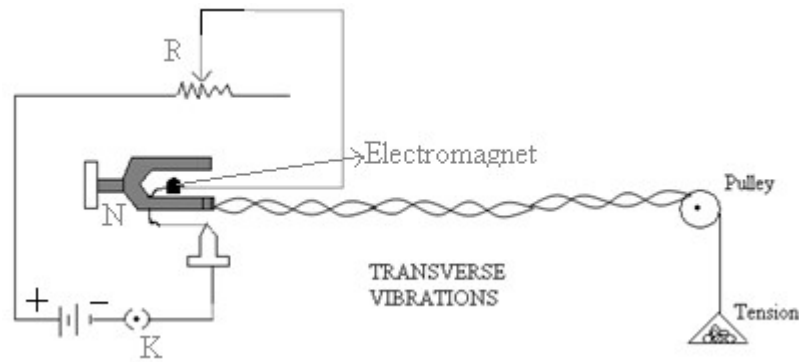
Thus string vibrates and forms many well defined loops. This is due to the stationary vibrations set up as results of superposition of the progressive waves from the pulley well defined loops are formed. When the frequency of each segment coincides with the frequency of the fork, the frequency of 'n' transverse vibrations of the stretched string by a tension of 'T' dyne is given by

$$n = \frac{p}{2l} \sqrt{\frac{T}{\mu}}$$

Where, T= Tension applied to the string = Mg

μ = mass per unit length of string

l = loop length of the string.



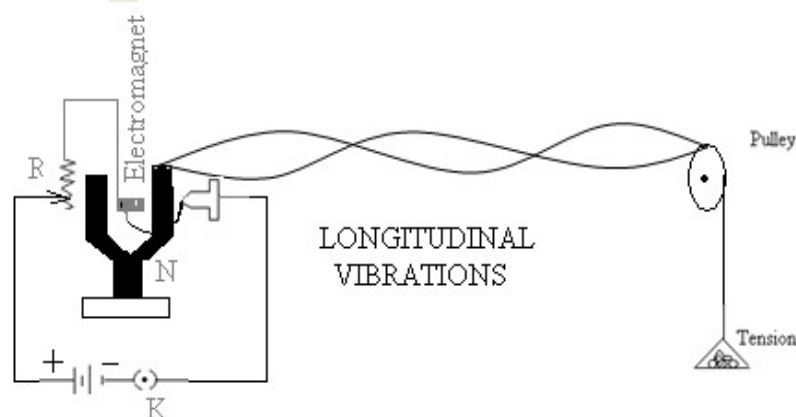
B. Longitudinal Arrangement:

When the direction of motion of the prong is along the length of the thread, the vibrations of the thread represent longitudinal mode of vibration.

When the fork is placed in the longitudinal position and the string makes longitudinal vibrations. The frequency of the stretched string will be half of the frequency 'n' of the tuning fork. That is when well defined loops are formed on the string.

The experiment is repeated by increasing (or) decreasing the load (m) so that the no. of loops increases (or) decreases by one. The experiment is repeated till the observation are recorded in table – I. Next, the table (or) vibrator is arranged for longitudinal vibrations the experiment is repeated for longitudinal vibrations and the observation are for recorded in table – II.

At the end of the experiment the mass ' m_p ' of the pan, the mass of the string and the length of the string is noted.



Observations:

Mass per unit length of the thread (μ) = _____ gm

Table – I**Transverse Arrangement:**

S. No.	Mass loaded to pan 'm' (g)	Mass of the pan 'm _p ' (g)	Total mass M =(m+m _p) (g)	No. of loops (x)	No. of loops (p)	Loop length l (cm)
1.						
2.						
3.						
4.						

Table – II:**Longitudinal Arrangement:**

S. No.	Mass loaded to pan 'm' (g)	Mass of the pan 'm _p ' (g)	Total mass M =(m+m _p) (g)	No. of loops (x)	No. of loops (p)
1.					
2.					
3.					
4.					

Calculations:

Dr. S. Damodar

Precautions:

1. A thin long and inelastic thread should be used.
2. The loops should be well defined and confined to a single plane.

Result:

The frequency of a vibrating tuning fork is measured by using Melde's experiment as

n = _____ Hz.